Research Studies Panel

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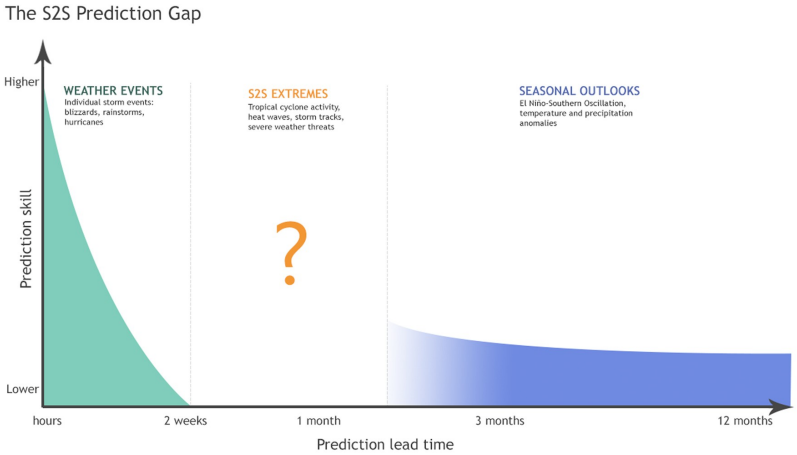
What Is the Prediction Gap?



Source: Mariotti et al. (2018)

24%

of all floods globally follow drought

Matanó et al. (2024)

What Does the Literature Say?

Literature operates at two levels

✓ Predict drought or flood separately

✓ Characterise joint behaviour historically

✗ Does met skill carry through to multi-hazard event skill?

Nobody has tested this

Multi-hazard event skill

Regime conditioning

Event detection

Impact variables

System selection

Research Arc

1. *How does prediction skill at S2S lead times compare across forecast variables, impact variables, and multi-hazard events?*
2. *Under what atmospheric conditions does this skill emerge?*

Paper 1
Drought-to-flood
temporally compounding

Paper 2
RES droughts
multivariate

Paper 3
ANTICIPATE COST Action
to be determined

Paper 1

Met skill	Impact skill	Event skill	Regime conditioning	ML
Verification	**Streamflow**	Detection	Atmospheric state	Added value?

Subseasonal vs Seasonal

Subseasonal (EEFH)

Newer physics
183 inits / year
11 members
↑ **Temporal sampling**

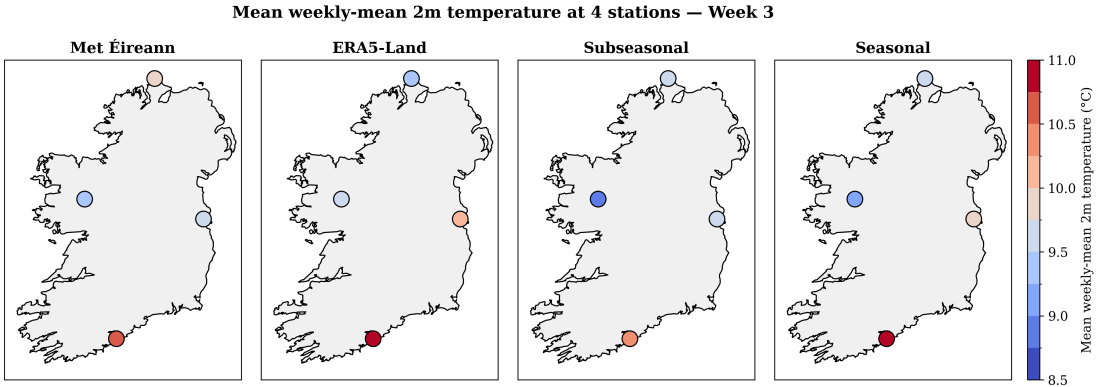
vs

Seasonal (SEAS5)

Older physics
12 inits / year
25 members
↑ **Ensemble sampling**

2006–2016 | 0.3° grid | t2m, tp, ws10

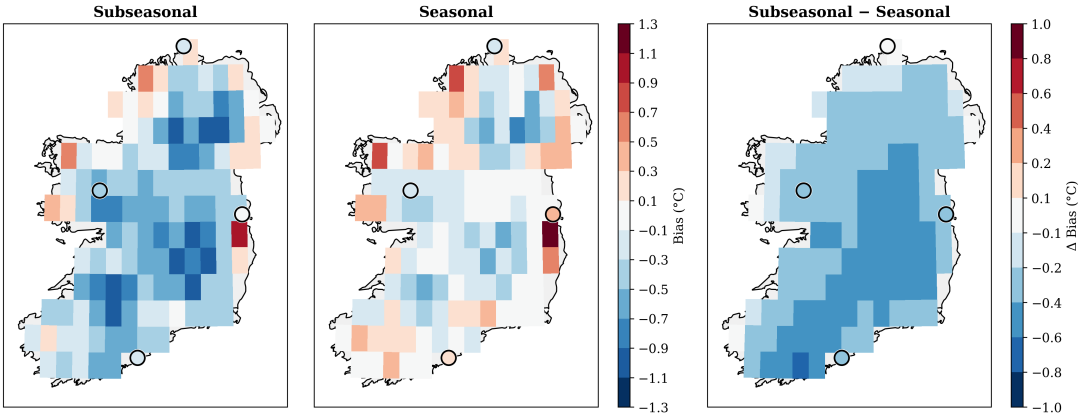
Observation–Forecast Temperature Chain



1. ERA5-Land warm-biased relative to stations

2. Seasonal warmer than subseasonal

Spatial Verification



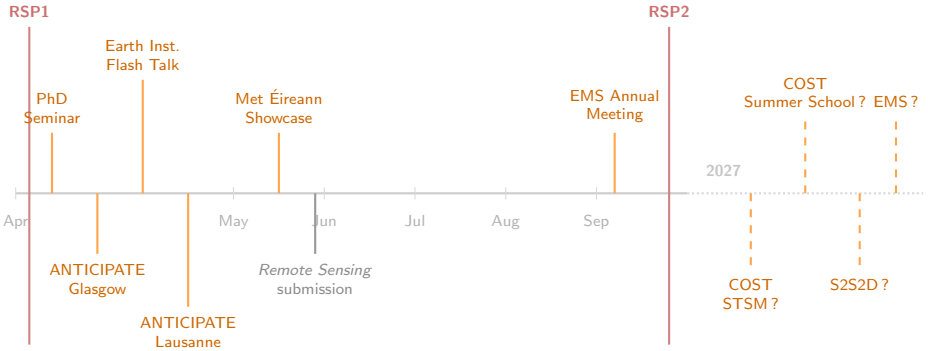
1. Subseasonal cold bias exaggerated by warm reference

2. Seasonal warm bias is genuine

Training & Development

Credits ACM41070 (5) | STAT41130 (5) | COMP41850 (5) | RPL (15)

Training Research integrity | HPC | Ark presentation | DestinE ML



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- Rashid, M. M., and T. Wahl, 2022: Hydrologic risk from consecutive dry and wet extremes at the global scale. *Environ. Res. Commun.*, **4**, 071001.
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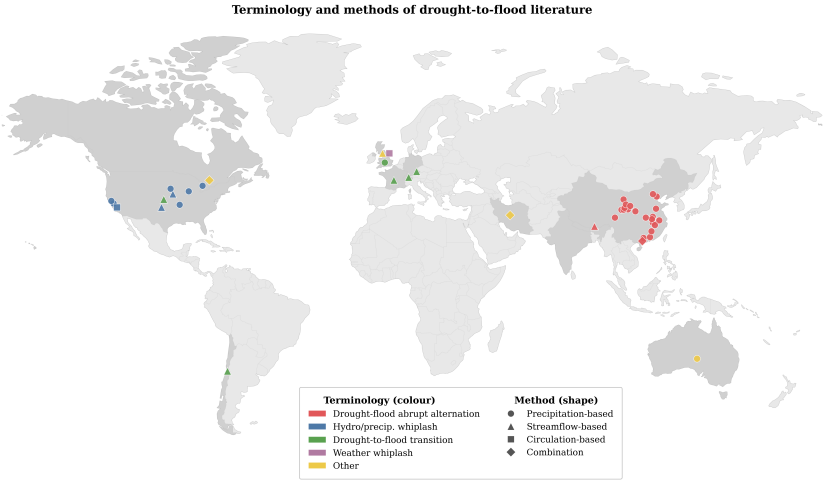
How Do You Detect These Events?



Joint indices

Compress the swing into one number

vs

Episode-linking

Detect separately, link if close in time

Three strategies for detecting transitions

1. Raw thresholding

Streamflow

↓

Threshold directly (e.g. <Q20, >Q90)

↓

Link drought + flood episodes by proximity

↓

DTF events

Anderson (2025), Götte & Brunner (2024), Yang (2025), Matanó (2024)

2. Index-based

Precipitation

↓

Compute index (SPI, SPEI, LDFAI, etc.)

↓

Threshold the index to identify events

↓

DTF events

Wang (2024), Bai (2024), Zhao (2024)

3. Regime classification

500 hPa height anomalies

↓

Cluster circulation patterns

↓

Identify regime shifts

↓

DTF events

Francis et al. 2023

Yang et al. (2025) applied methods 1 and 2 to 7 decades of US catchment data and found **opposite frequency trends**.

Dataset Overview

Forecast systems

System	Source	IFS cycle	Grid	Members	Reforecast period	Lead range
Seasonal (SEAS5)	MARS API	Cy43r1	0.28°	25	1981–2016	215 days
Subseasonal (EEFH)	MARS API	Cy49r1	0.28°	11	2005–2024	46 days

Reference datasets

Dataset	Source	Grid	Coverage
ERA5-Land	MARS API	0.1°	1950–present
Met Éireann	Met Éireann archive	Station	1945–present (varies by station)

Common analysis window: 2006–2016 **Climatology:** 1991–2020 (WMO) **Output grid:** 0.3°
9 Tier 1 stations: full climatology + full analysis coverage **Variables:** t2m, tp, ws10

Subseasonal vs Seasonal: System Comparison

	Subseasonal (EEFH)	**Seasonal (SEAS5)**
IFS cycle	Cy49r1 (2024)	Cy43r1 (2016)
Grid type	Cubic octahedral	Reduced Gaussian
Resolution	∼32 km (TCo319)	∼36 km (T319)
Vertical levels	137	91
Stochastic physics	SPP (27 parameters)	SPPT3 + SKEB
Ozone–radiation	Fully interactive	Climatological
Reforecast members	11	25
Real-time members	101	51
Init frequency	Every odd day	1st of month
Reforecast period	Prior 20 yr (rolling)	1981–2016 (fixed)
Lead range	46 days	215 days
Atm. initial conds	ERA5	ERA-Interim

Three research traditions

	DFAA	**Western whiplash / transition**	**Weather whiplash**
Origin	Chinese hydrology	US and Europe	Atmospheric science
Primary variable	Precipitation (SPI-based)	Precipitation, streamflow, or both	500 hPa geopotential heigh
Detection logic	Purpose-built alternation indices (single scalar)	Independent episode detection, linked by temporal proximity	Regime classification via self-organising maps
Seasonality	Monsoon-specific	Various	Synoptic timescales
Research focus	Characterisation	Trends and prediction	Regime dynamics
Role of ML	None	Emerging	None

DFAA indices embed weighting coefficients calibrated for monsoon basins — **no study has re-calibrated them for Atlantic Europe.**

How are drought-to-flood events defined?

Study	Region	Strategy	Drought def.	Flood def.	Transition rule
Götte & Brunner (2024)	US	1	Streamflow $<$ Q20 (varies by day-of-year)	Streamflow $>$ Q50 of annual maxima	$\geq$10 days
Yang et al. (2025)	US	1	Streamflow $<$ Q50 of annual 7-day min	Streamflow $>$ Q50 of annual maxima	$<$90 days
Matanó et al. (2024)	Global	1	$<$Q15 in $\geq$1 hydrological variable	Streamflow $>$ Q85	$<$30 days
Guimarães et al. (2026)	France	1	Streamflow $<$ Q20	Streamflow $>$ Q99	No gap
Wang et al. (2024)	China	2	SPI $\rightarrow$ MSDFAI (alternation index)	SPI $\rightarrow$ MSDFAI (alternation index)	Index $\geq$ 1

Anderson et al. (2025) compared multiple threshold types on 8 catchments with known transitions and detected only **3 of 8**. **No study has tested any approach on Irish catchments.**

Verification Metrics

f_i = forecast weekly mean, o_i = observed weekly mean, $\bar{c}_i$ = climatological mean (1991–2020)

$f'_i = f_i - \bar{c}_i$ = forecast anomaly, $o'_i = o_i - \bar{c}_i$ = observed anomaly, N = number of cases

$$\text{Bias} = \frac{1}{N}\sum_{i=1}^{N}(f_i - o_i)$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(f_i - o_i)^2}$$

$$\text{ACC} = \frac{\sum_i f'_i\, o'_i}{\sqrt{\sum_i (f'_i)^2}\,\sqrt{\sum_i (o'_i)^2}}$$

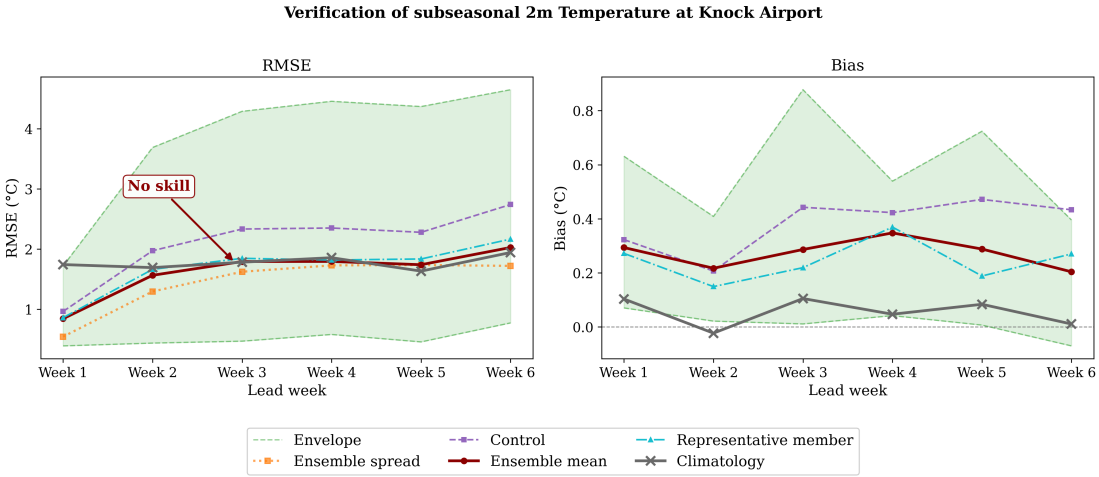
Uncentred, unweighted, matching the ECMWF operational definition. Computed as spatial pattern correlation per init (across gridpoints), then averaged across inits.

Pipeline Variables

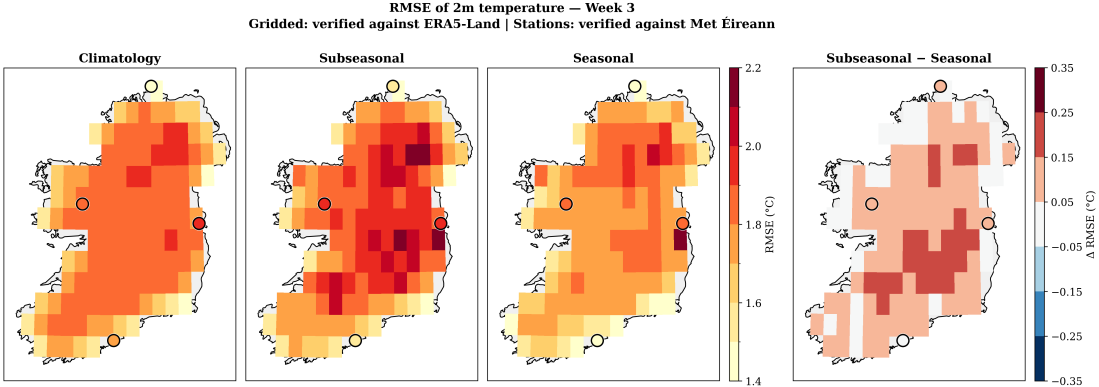
Var.	Seasonal	Subseasonal	ERA5-Land	Met Éireann	Verif. res.
t2m	Daily mean (from 6h inst.)	Daily mean (from 6h inst.)	Daily mean (from 6h subset)	Daily mean (from hourly)	Daily
tp	Daily total (de-accum. 6h)	Daily total (de-accum. 6h)	Daily total (from hourly)	Daily total (native)	Daily
ws10	Daily mean (from 6h u10, v10)	Daily mean (from 6h u10, v10)	Daily mean (from 6h u10, v10)	Daily mean (from hourly)	Daily

All variables aggregated to **weekly means** for scoring.

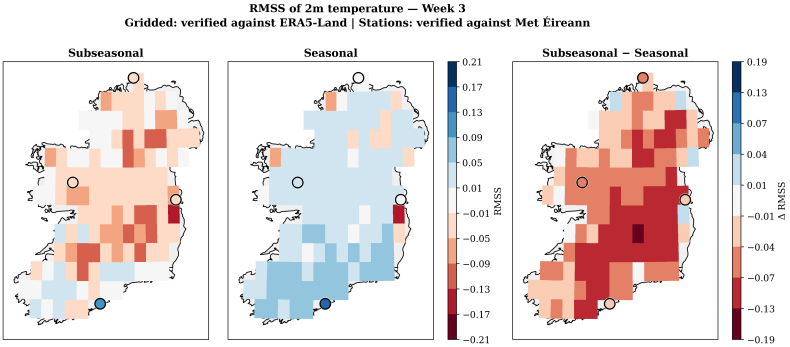
Station-Level Verification



Spatial RMSE



Spatial Skill Score



RMSS

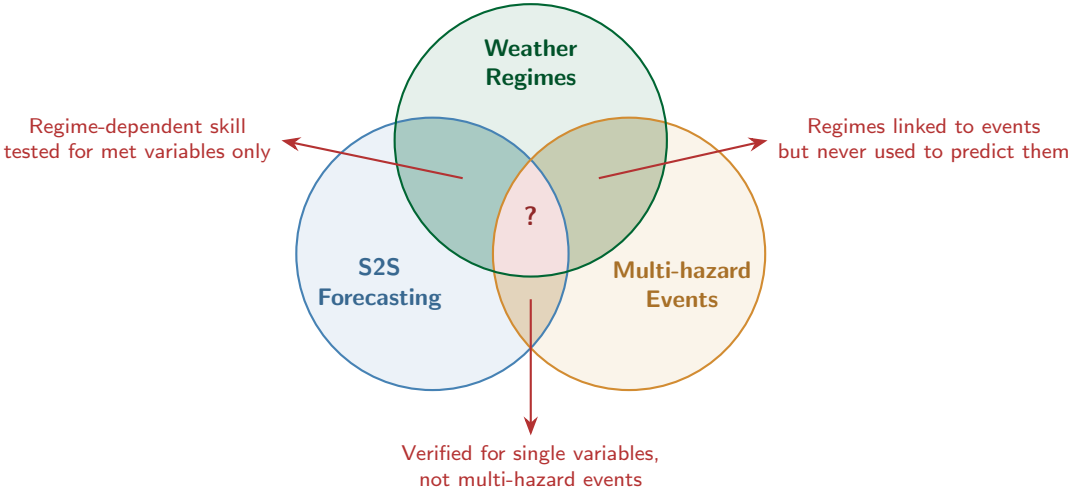
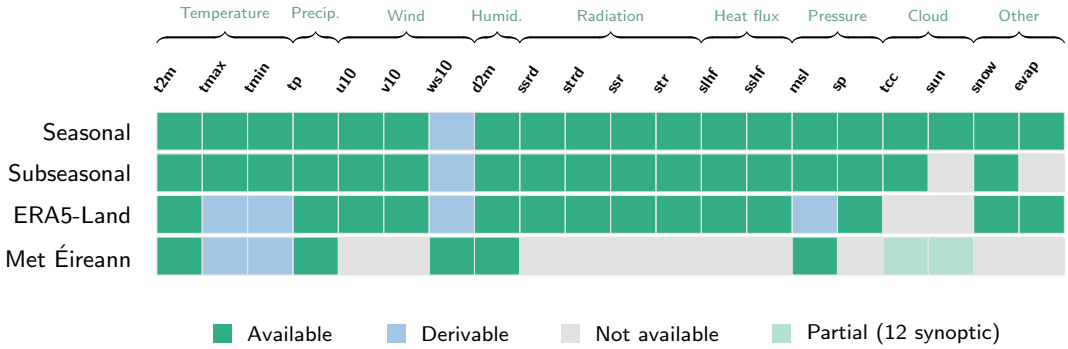
$$\text{RMSS} = 1 - \frac{\text{RMSE}_{\text{fc}}}{\text{RMSE}_{\text{clim}}}$$

RMSS > 0: forecast beats climatology

RMSS $= 0$: no skill

RMSS < 0: worse than climatology

Where the Literature Review Sits



Variable Availability